



Constraining Geothermal Flux In Antarctica Using An Ice-and-Heat Flow Model



Sirray Smith¹, T.J. Fudge²

Department of Physics and Engineering Physics at Morgan State University¹ Department of Earth and Space Sciences at University of Washington²

1. Introduction

Geothermal Flux is the rate of heat coming from the interior of the Earth. This heat comes from two primary sources:

- **Primordial Heat**
- **Heat Producing Elements**

Geothermal Flux helps determine if the ice sheet is frozen to the bed.

- Ice sheets that are frozen to the bed flow faster than ice sheets that are not grounded.

Raymond Arches help determine if an ice frozen to the bed.

- **Raymond Arches** form at ice divides.
- These can only be formed if the ice sheet is frozen to the bed due to the deformity of the ice after thousands of years.

Geothermal Flux is one of the factors for **basal melting**

- This causes sliding increasing the flow of ice.

2. Study Area



Figure 1: Map of Antarctica

Geothermal Flux is poorly constrained. Using data from Antarctica, primarily **West Antarctica** to find the max GTF. Using the characteristics of **ice rises**.

- **Ice Rises** are dome-shaped mass of ice surrounded either by an ice shelf or partly by an ice shelf.
- Areas circled are the main ice rises used.

3. Methods: Modeling

A Model was used to find the **Maximum Geothermal Flux Constraint**. The model allowed for the input of the ice rise characteristics.

- The **ice thickness**, **accumulation rate**, and **surface temperature** were inputted to see the **Maximum Geothermal Flux** that would cause **basal melt**.

```
while melt_history(end) == 0 ;
q_geo = q_geo0 + c

[calc_times, H_vec, Hdot_vec, h_vec, slope, w_vec, TEMP_climate, bdot,...
q_geo_unused, mdot, particle_times, phi_B ] ...
= engelhardt_setup_many(vals{j,:}) ;
```

- The model would take the parameters of the ice rise and an initial geothermal flux to begin.
- The model is using these factors to find the internal temperature of the ice to see when basal melt will begin on the ice rise.

```
[ T_z, RHO, Z, melt_history, melt_times, T_base, TEMP_store, Z_store, ...
particle_times, z_particle, Z_P, s_const, ...
T_particle, T_particle_matrix, z_particle_matrix, TIME_vec ] ...
= transientTemp( calc_times, H_vec, h_vec, slope, TEMP_surft,...
bdot, c_vals, k_vals, VD_vals, w_vec, mdot, ...
u_s, du_s_dx, w, dw_dx, shape_params, xpos, particle_times, phi_B );
c=c+.001 ;
figure(1)
hold on;
plot(T_z,Z_P, 'DisplayName',sprintf('q_geo = %.3f',q_geo));
ylim([0 H_vec(end)]);
end
```

4. Results

Names	Latitude	Longitude	Ice Thickness (m)	Surface Temperature (°C)	Accumulation Rate (m/a)	Max Geothermal Flux mW/m ²	Basal Temperature (°C)	Geothermal Flux (mW/m ²)
Siple Dome	-81.85	-148.81	1,004	-25.5	0.13	78	2.54	72
Engelhardt Ridge	-83.33	-148.00	850	-25.5	0.13	87	N/A	N/A
Roosevelt Island	-79.42	-162.00	765	-24	0.22	101	-4.3	85
Shabla Ridge	-80.50	-140.00	1,000	-25	0.13	76	N/A	N/A
Fletcher Promontory	-78.42	-80.00	854	-27.1	0.38	153	-18	63
Crary Ice Rise	-82.83	-172.50	550	-27	0.44	190	N/A	N/A
Berkner Island	-79.50	-47.50	948	-26.5	0.13	84	-11.6	50
Kealeley Ice Rise	-77.00	-83.00	450	-30	0.44	228	N/A	N/A
Skytrain Ice Rise	-79.67	-78.50	651	-25.9	0.14	109	-14.9	59
Halvfaryrgen Ice Rise	-71.16	-6.70	750	-17.9	1.2	144	N/A	N/A
James Ross Island	-64.15	-57.75	363	-14.4	0.63	148	-8.5	72

Table 1: Shows the Ice Rise and each of the characteristics, along with the Geothermal Flux.

Table 1 shows each of the complete Ice Rises found in West Antarctica. It includes each of the **thickness** of the varying ice sheets along with the **surface temperature** and **accumulation rate**. These were used to find the **max geothermal flux** along with the geothermal flux is a basal temperature is listed.

How the Model Works

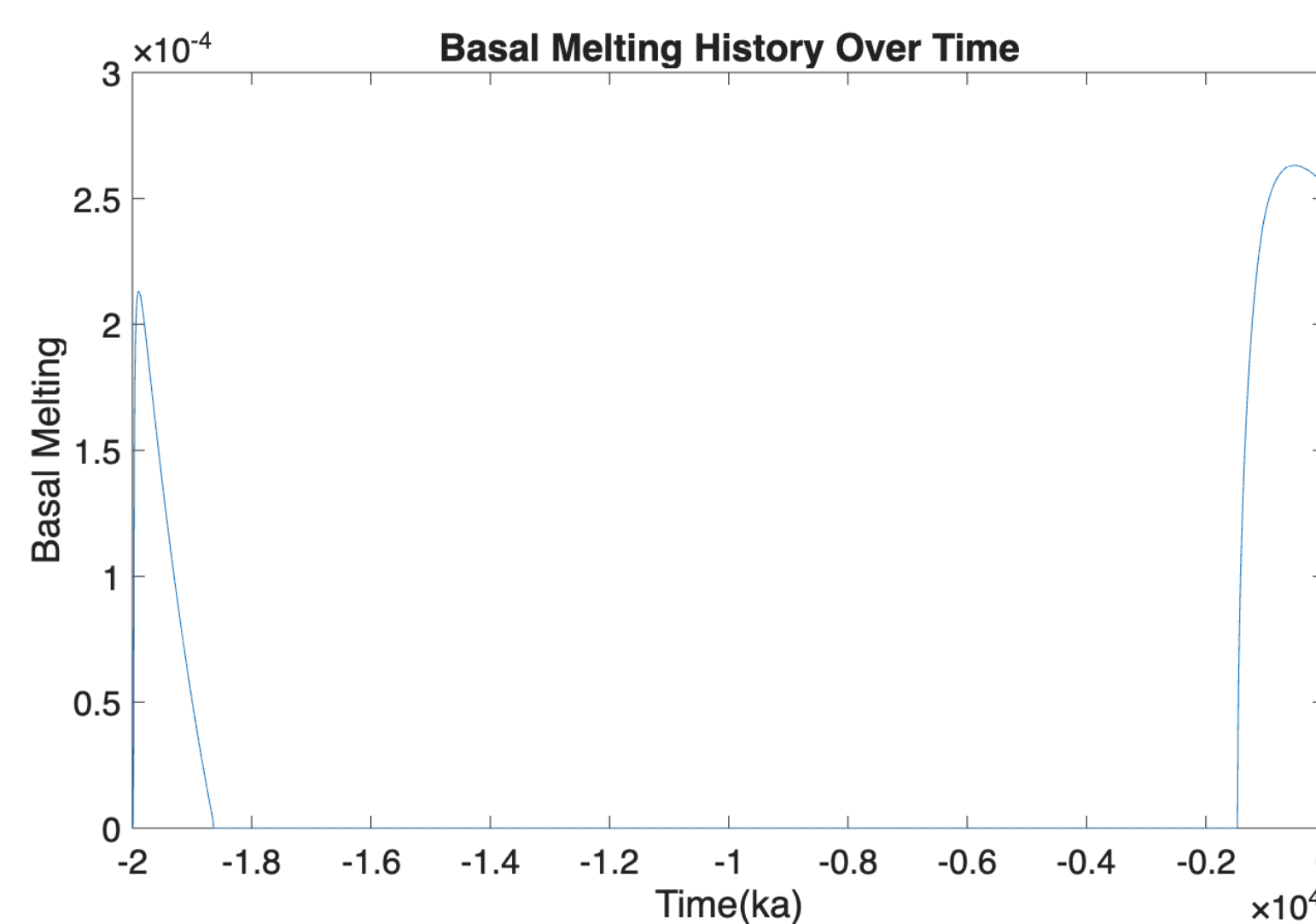


Figure 2: Basal Melt over time for Siple Dome, West Antarctica

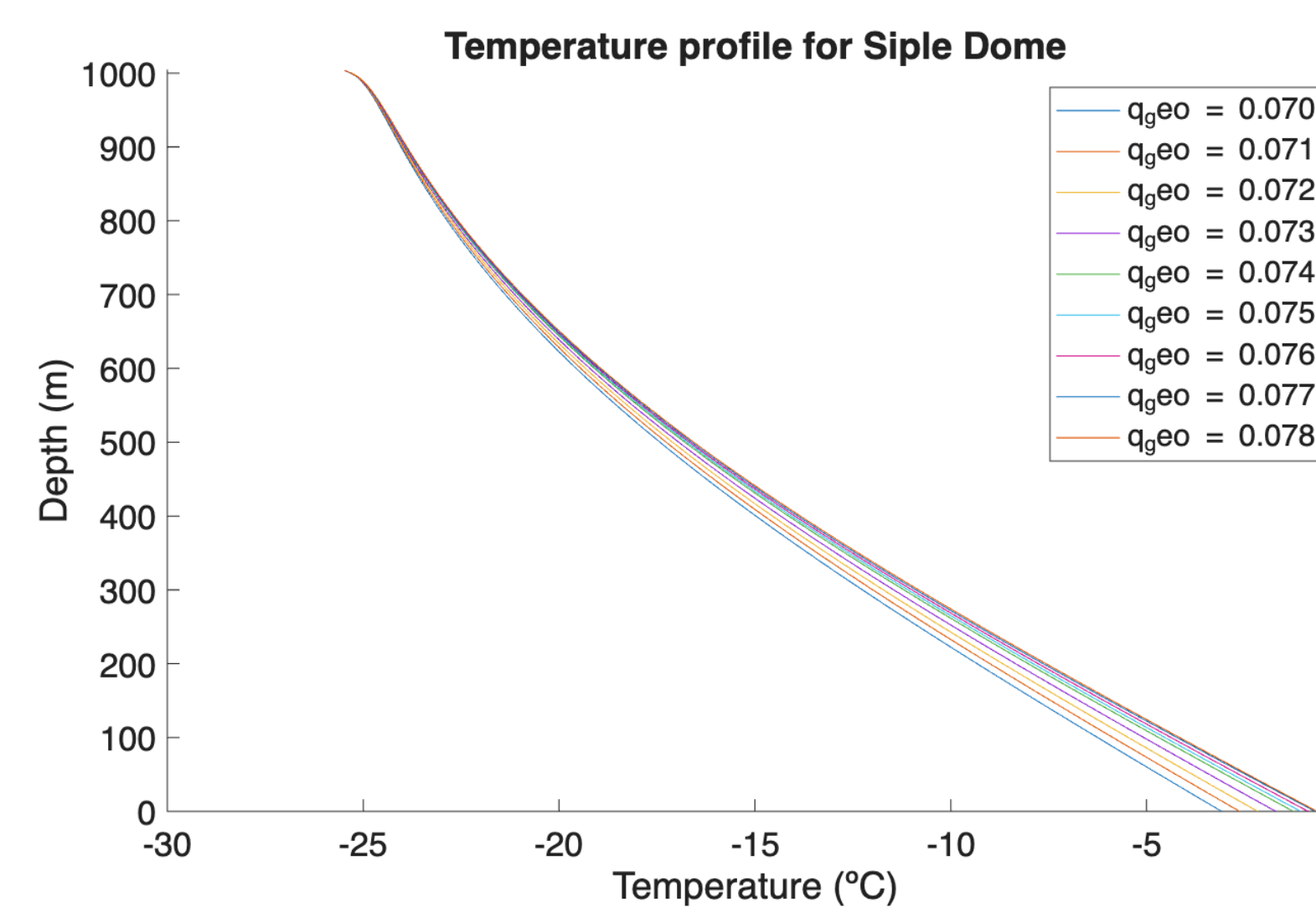
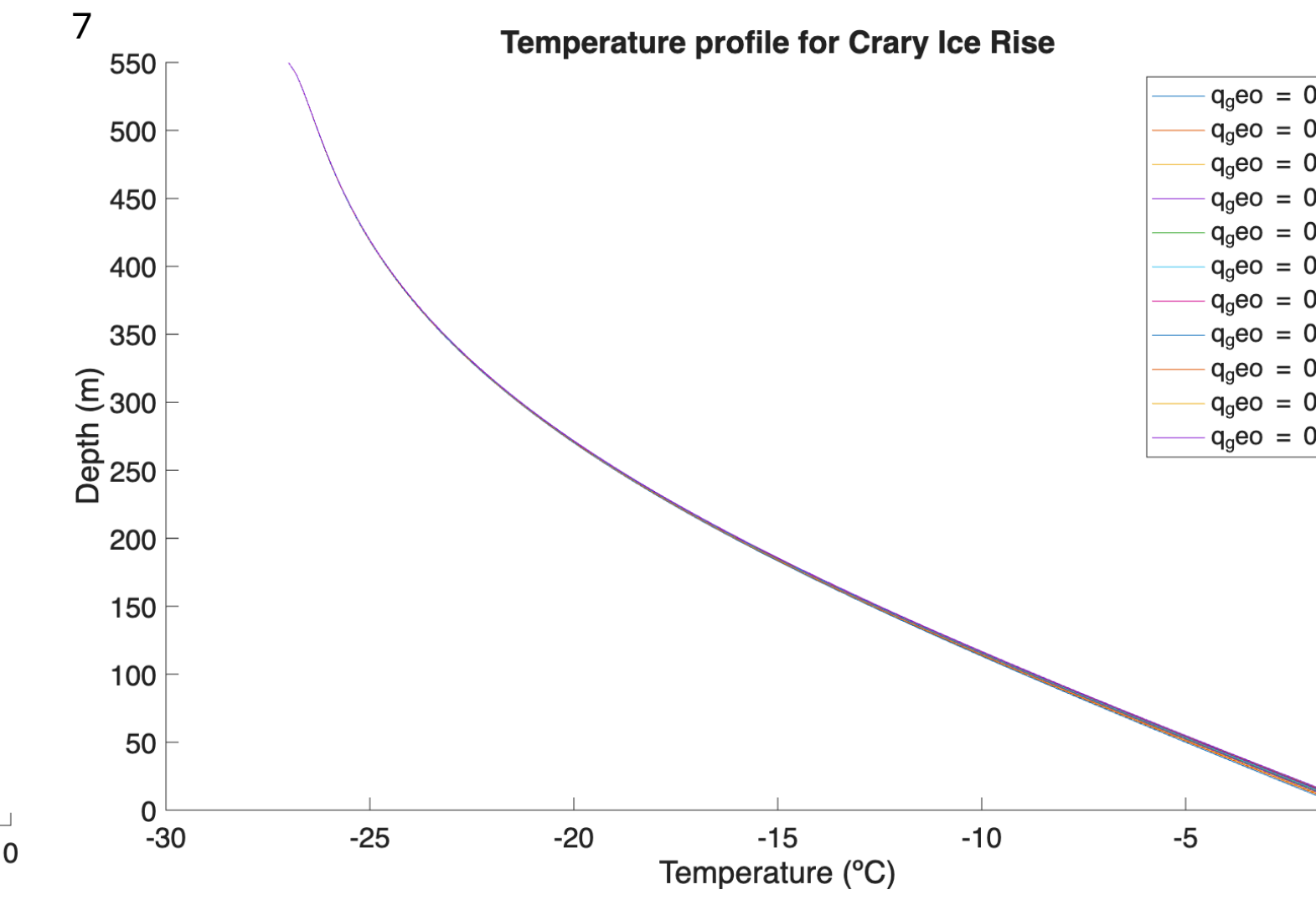
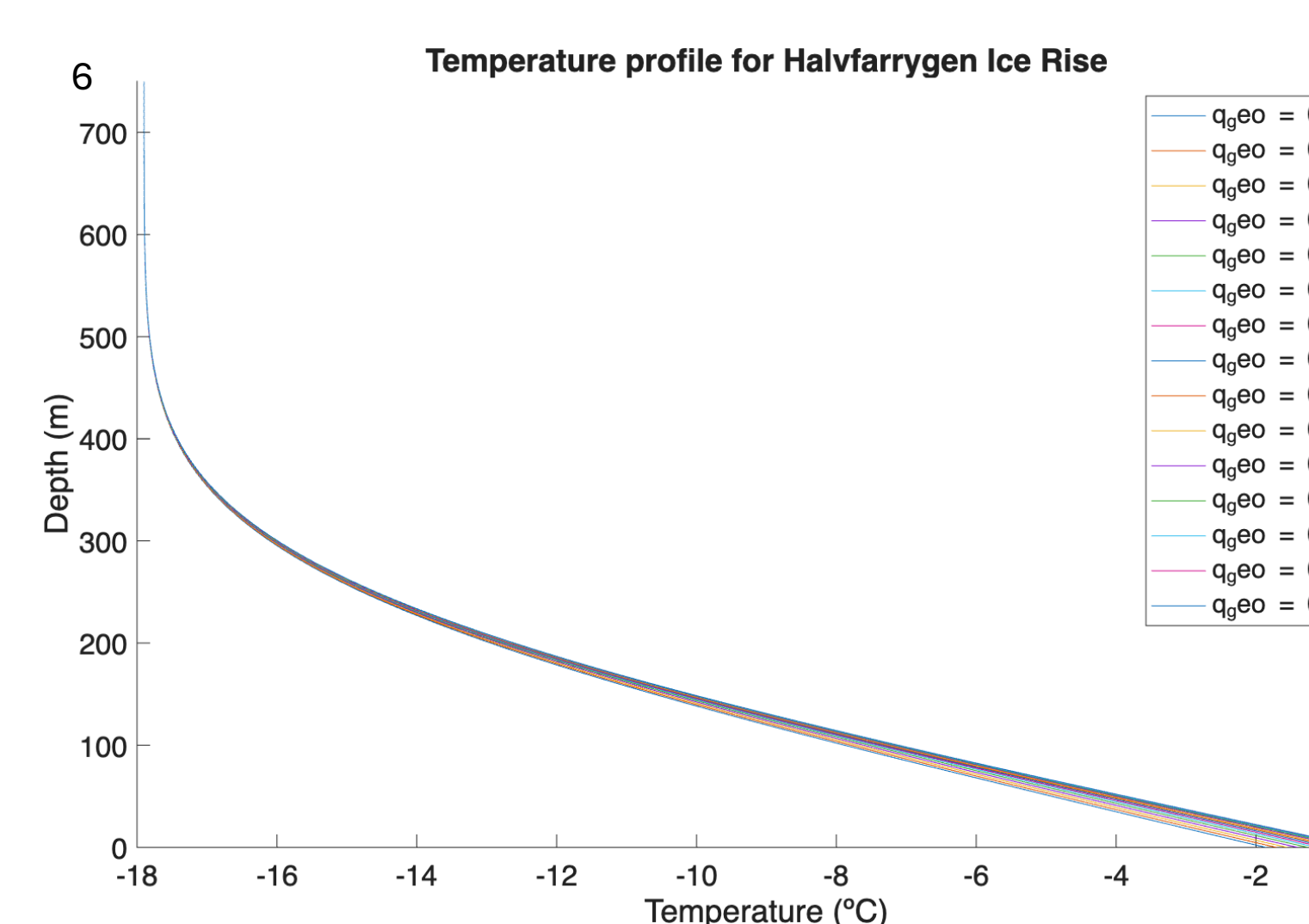
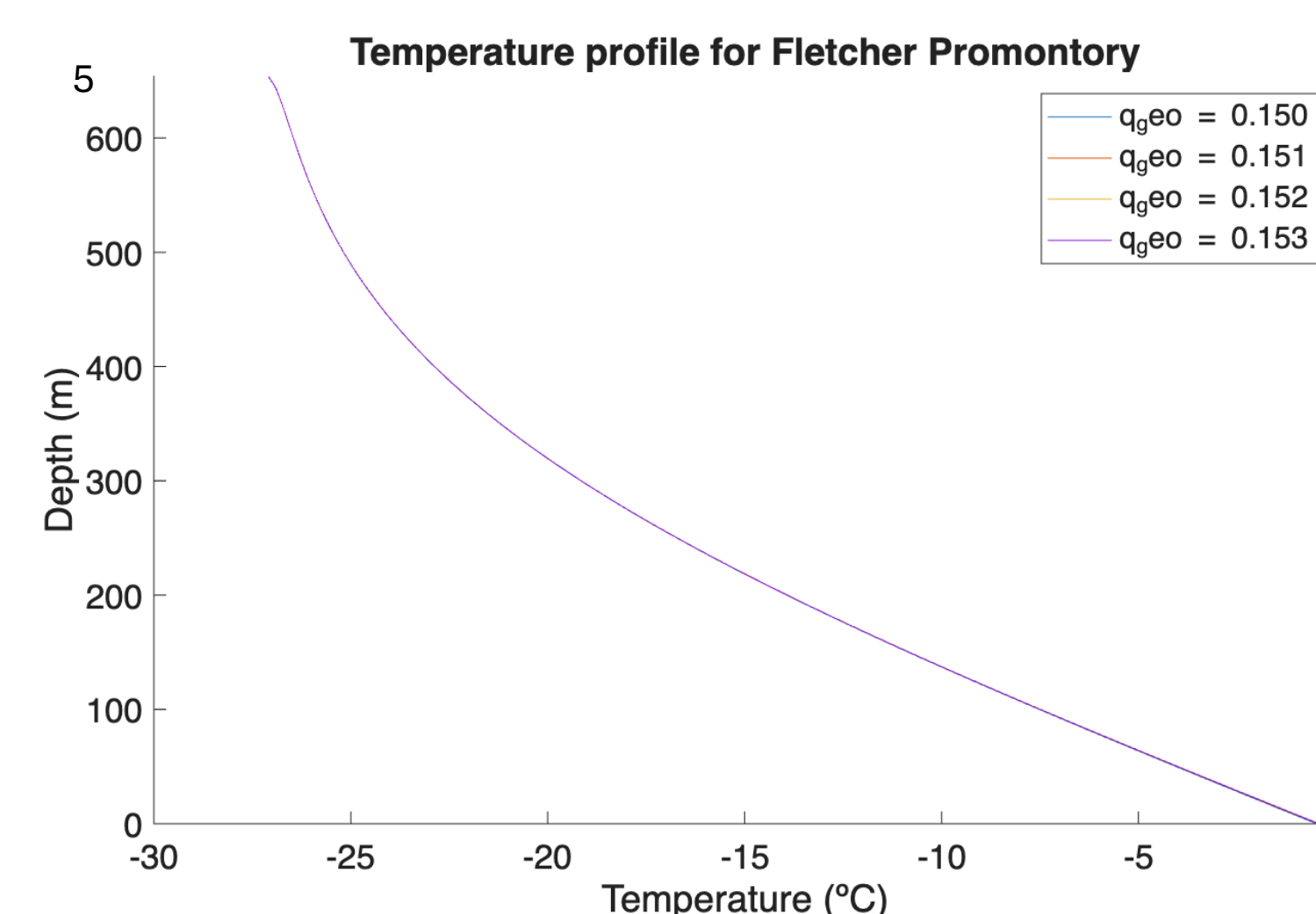
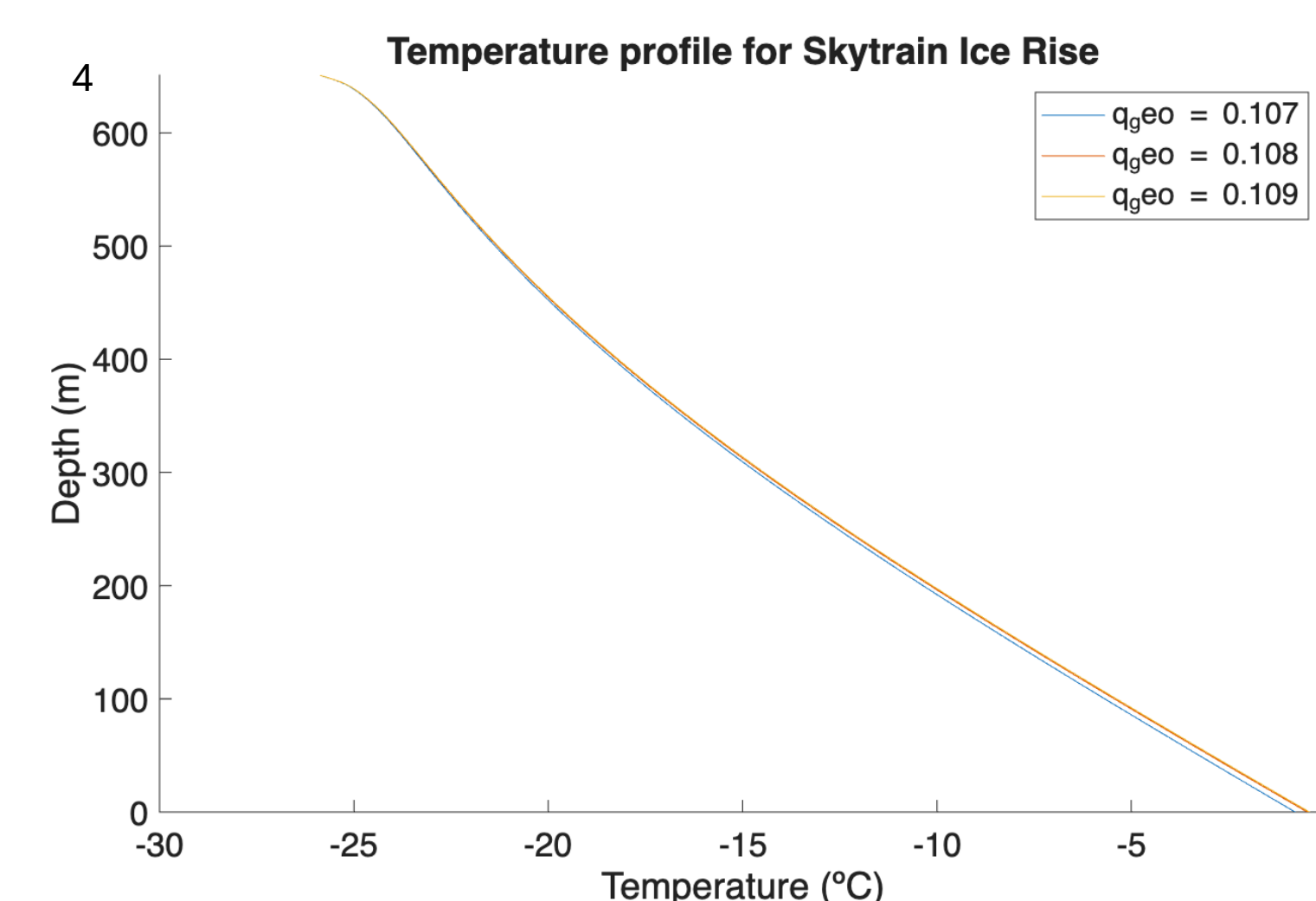


Figure 3: Temperature Profile for Siple Dome, West Antarctica

Figures 2 and 3 show the basal melt and temperature profile of the Siple Dome ice rise located in West Antarctica. **Figure 2** explains how the basal melt works in the model. The basal melt is equal to zero and spikes when the **maximum geothermal flux** is found. **Figure 3** shows the temperature profile that shows the progression of the **geothermal flux** until the maximum is found.

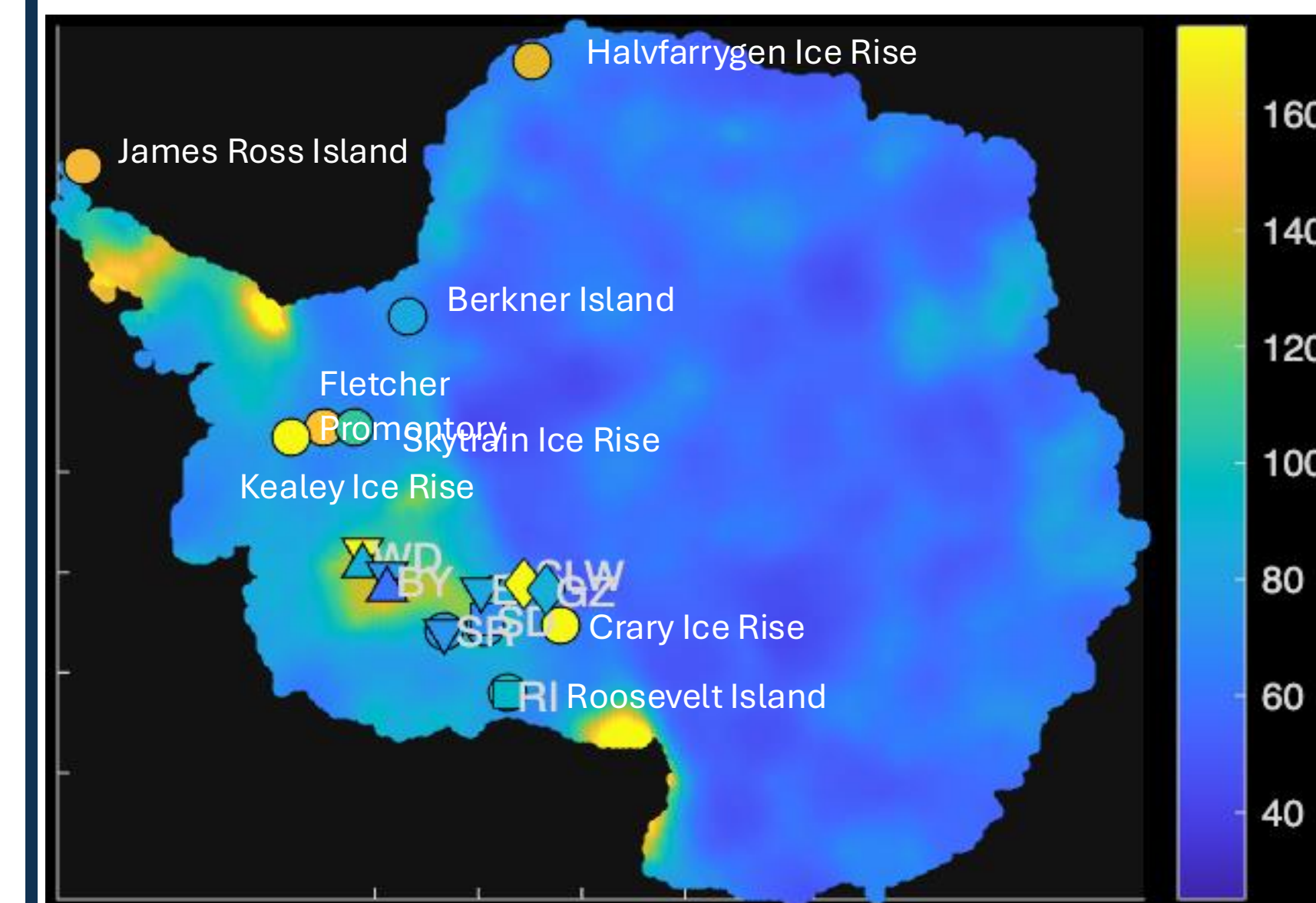
More Temperature Profiles



Figures 4-7: The Temperature Profiles for the Skytrain Ice Rise, Fletcher Promontory, Halvfaryrgen Ice Rise, and Crary Ice Rise located in West Antarctica.

Figure 4-7 shows the different temperature profiles for the different ice rises in West Antarctica. The Skytrain Ice and and the Fletcher Promontory both have known basal temperatures. The basal temperature for each is -14.9 °C and -18 °C. The model continues until a **max geothermal flux** is reached and does not account for basal temperature. Due to knowing this factor, the **geothermal flux** at those temperatures have to be located. For the Skytrain Ice rise the **geothermal flux** is 59 mW/m² instead of the **max** of 109 mW/m². For Fletcher Promontory the **geothermal flux** is 63mW/m² instead of the **max** of 153 mW/m².

5. Discussion



Using Martos et al. was able to compare current **max geothermal flux** to past results.

- The plotted **ice rises** seem to be higher than the maps.
 - Possibly due to plotting the **maximum geothermal flux**.
- Berkner Island has a lower **geothermal flux** compared to the surrounding area when taking in account the basal temperature.
 - If model was able to take in account the basal temperature, mapped plot points may match the Martos Data and show key differences such as this one.

6. Conclusion

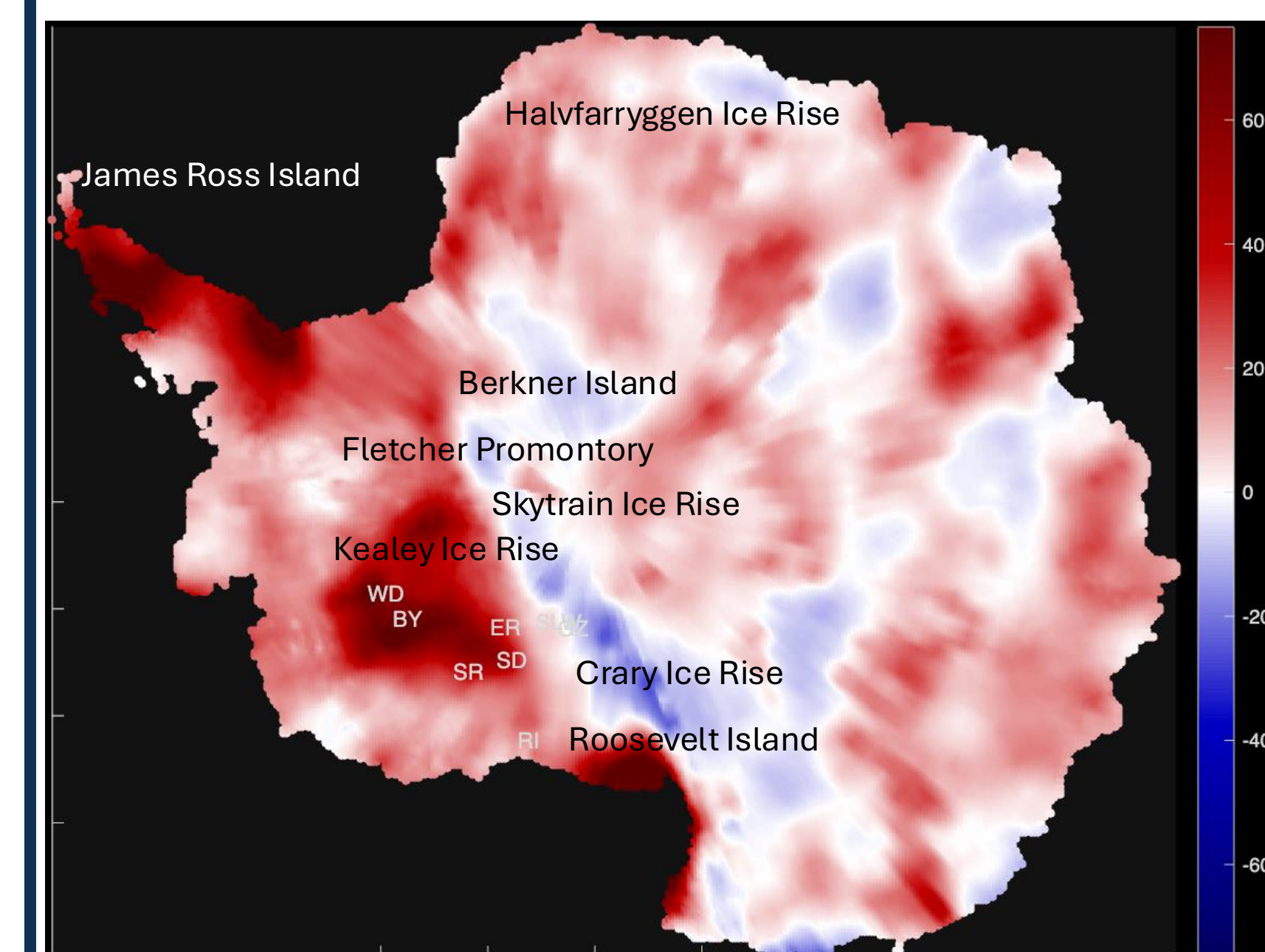


Figure 9: Comparing the difference in geothermal flux

Using Martos et al. was able to compare the differences in the geothermal flux.

- The color bar explains the differences between the two datasets.
 - **Red** meaning higher than the map and **blue** being lower.
 - Kealeley Ice Rise and the surrounding ice rises show to have a difference of around 20 mW/m².
- Future Work**
- To find information on ice rise in East Antarctica.
 - To find more information on ice rises in West Antarctica.

7. References & Acknowledgments

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3. Martos, Yasmina M., Manuel Catalán, Tom A. Jordan, et al. "Heat Flux Distribution of Antarctica Unveiled." *Geophysical Research Letters* 44, no. 22 (2017): 11,417–11,426. <https://doi.org/10.1002/2017GL075609>.

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